

Al-Nahrain University
College of Engineering
Department of Computer
Fourth Stage/First Semester



Control Engineering I
Experiment No.2: Closed Loop Control and Feedback Polarity

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Objectives:

To learn the basic configuration of a closed loop system and how it functions. To learn that the polarity of the feedback in a closed loop system must be negative for the system to work correctly To learn that care must be taken over polarities in order to ensure that when the setting value is changed, then the output changes in the appropriate direction

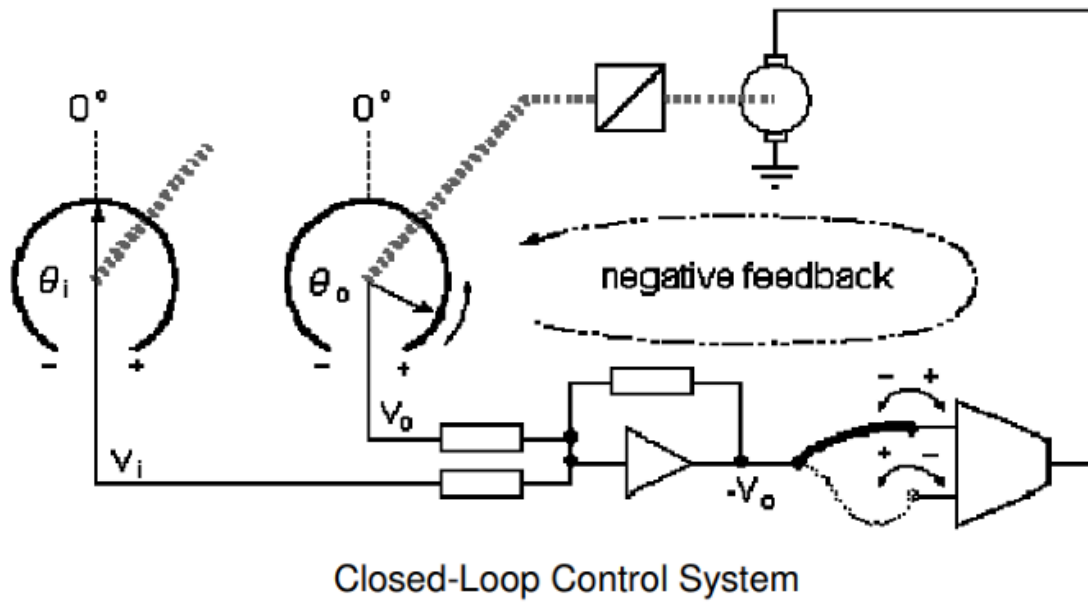
Procedure:

The Control & Instrumentation Principles 33-033 application was opened on the computer, and closed loop control and feedback polarity were selected.

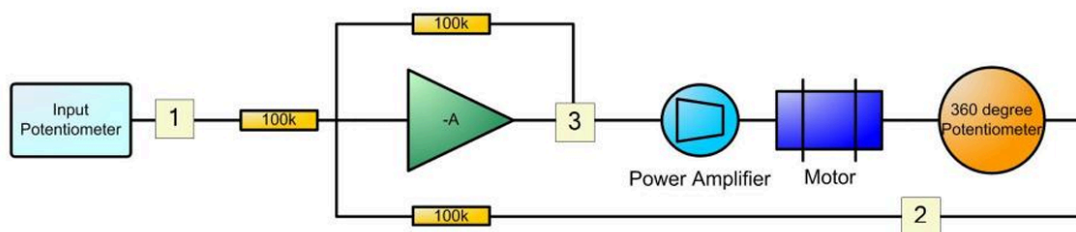
Practical 1: A Simple Closed Loop System

In a control system with input and output shafts, some method of measuring the input and output shaft angles and determining the difference between them was required. The error was then used to produce a voltage, or was measured directly as a voltage suitable for driving the power amplifier.

A convenient method to measure the shaft angles electrically was provided by attaching a potentiometer to each shaft, as shown in the diagram. The signals V_i and V_o were then combined in an operational amplifier so that an error signal was produced to operate the power amplifier.



The connections of the following block diagram were made, and the instructions of Practical 1 were followed.



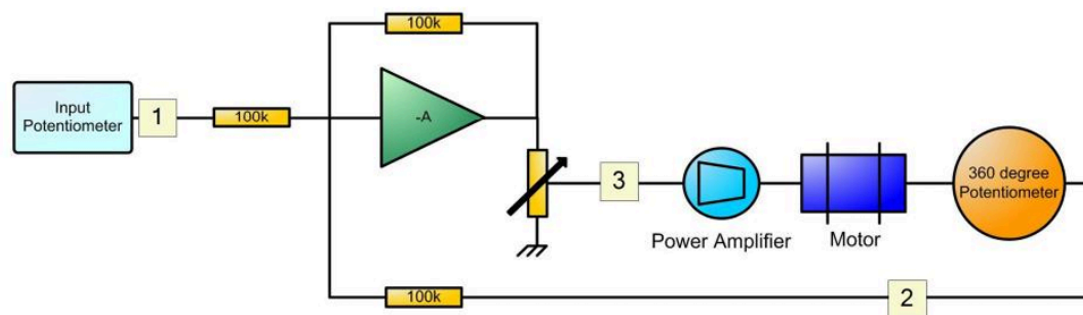
Practical 2: Feedback Polarity

The effect of negative closed-loop feedback on a system was demonstrated in Practical 1; in this practical, the effect of positive feedback on the same system is shown.

It is shown that for correct system operation, it is required that the error signal be rotated in the appropriate direction so that the motor is driven to reduce the error – this condition is known as negative feedback.

If the error signal is caused to rotate the motor in a way that increases the error, then positive feedback is produced, and the system is rendered useless.

The connections of the following block diagram were made, and the instructions of Practical 2 were followed.



Discussion:

Q1/ Explain the difference between practical 1 and 2?

-Practical 1 deals with negative feedback.

The system senses the difference between the input angle and the output angle. This error signal makes the motor rotate in the correct direction to reduce the error. Because the motor corrects itself, the system becomes stable and behaves the way a closed-loop system should.

Practical 2 deals with positive feedback.

Here, the polarity is reversed. The error signal makes the motor rotate in the wrong direction, which increases the error instead of reducing it. The system becomes unstable and cannot function correctly because any small error keeps growing.